

GOOD AND INCREASING DECOMPOSITIONS OF PERMUTATIONS

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1. DEFINITIONS

Let $n \geq 1$ and let $[n]$ denote $\{1, \dots, n\}$. For $i, j \in [n]$, let τ_{ij} denote the transposition swapping i and j - we call i and j the *endpoints* of τ_{ij} . For a collection of transpositions C , let $S_C = \{i \mid \tau_{ij} \in C \text{ for some } j \text{ and } i < j\}$ and $L_C = \{j \mid \tau_{ij} \in C \text{ for some } i \text{ and } i < j\}$. In other words, S and L are the sets of small and large endpoints occurring in C , respectively. We say a collection of transpositions C is *good* if $S_C \cap L_C = \emptyset$. For a permutation π , let

$$\text{inv}(\pi) = \{(i, j) \mid i < j \text{ and } \pi(i) > \pi(j)\},$$

and $\ell(\pi)$ denote the length of π (i.e. $\ell(\pi) = \#\text{inv}(\pi)$).

2. PROBLEM

For every permutation π , there is a good collection of transpositions that can be ordered as $\tau_1, \dots, \tau_k$ so that $\pi = \tau_1 \circ \dots \circ \tau_k$ and the sequence of lengths $\ell(\tau_1) < \ell(\tau_1 \circ \tau_2) < \dots < \ell(\tau_1 \circ \dots \circ \tau_k)$ is strictly increasing.

3. SOLUTION

For an n -permutation π , define

$$S(\pi) = \{i \in [n] \mid \pi(i) > i\} \text{ and } L(\pi) = \{i \in [n] \mid \pi(i) < i\}.$$

We will prove the following stronger statement.

Theorem 3.1. *For any n -permutation π that's not the identity, there is a good collection $C = \{\tau_i\}_{i=1}^k$ that can be ordered so that $\pi = \tau_1 \circ \dots \circ \tau_k$ and the sequence of lengths $\ell(\tau_1) < \ell(\tau_1 \circ \tau_2) < \dots < \ell(\tau_1 \circ \dots \circ \tau_k)$ is strictly increasing, and such that $S_C = S(\pi)$ and $L_C = L(\pi)$.*

Before we can prove this statement, we need the following lemmas.

Lemma 3.2. *Let π be an n -permutation that's not the identity function (hence trivially $L(\pi) \neq \emptyset$). Let $j = \max L(\pi)$. If $i \leq j$ then $\pi(i) \leq j$.*

Proof. Suppose that $\pi(i) > j$. Define the sequence $(z_t)_{t \geq 0}$ as $z_0 = i$ and $z_{t+1} = \pi(z_t)$. It's clear that $z_1 = \pi(i) > j$. Suppose $z_t > j$ for some $t \geq 1$. Then z_t is larger than any element in $L(\pi)$, and so $z_{t+1} = \pi(z_t) \geq z_t > j$. Hence, inductively, $z_t > j$ for all $t \geq 1$. However, the orbit of z_0 under π is finite, and so $i = z_0 = z_t > j$ for some $t \geq 1$, which contradicts our initial assumption that $i \leq j$. $\square$

Lemma 3.3. *Let π be an n -permutation that's not the identity function. Then there are i, j such that $i \leq \pi(j) < \pi(i) \leq j$ and $\ell(\pi^*) < \ell(\pi)$ for $\pi^* = \pi \circ \tau_{ij}$.*

Proof. Since π is not the identity, $L(\pi) \neq \emptyset$. Let $j = \max L(\pi)$. Note that $\pi(j) < j$ since $j \in L(\pi)$, and so $j \notin [1, \pi(j)]$ and $\pi(j) \in [1, \pi(j)]$. If $1 \leq t \leq \pi(j)$ for all $1 \leq t \leq \pi(j)$ then π restrict to a permutation of $[1, \pi(j)]$, but that cannot be the case since $j \notin [1, \pi(j)]$ and $\pi(j) \in [1, \pi(j)]$. Hence, there must be some t such that $1 \leq t \leq \pi(j) < \pi(t)$. Note that $t \leq \pi(j) < j$, and so, by Lemma 3.2, we have

$$1 \leq t \leq \pi(j) < \pi(t) \leq j.$$

Let $i = \min\{t \mid 1 \leq t \leq \pi(j) < \pi(t) \leq j\}$.

Let $\pi^* = \pi \circ \tau_{ij}$. Note that (i, j) is in $\text{inv}(\pi)$ but not in $\text{inv}(\pi^*)$. Any element that is not in $\text{inv}(\pi)$ but is in $\text{inv}(\pi^*)$ must be of the form (p, i) , (i, p) , (j, p) , (p, j) for some $p \notin \{i, j\}$.

Suppose $i < p$. Then $(i, p) \notin \text{inv}(\pi)$ implies $\pi(p) > \pi(i)$ and $(i, p) \in \text{inv}(\pi^*)$ implies $\pi(j) > \pi(p)$. This would imply $\pi(i) < \pi(p) < \pi(j)$ contradicting the fact that $\pi(j) < \pi(i)$.

Suppose $p < j$. Then $(p, j) \notin \text{inv}(\pi)$ implies $\pi(j) > \pi(p)$ and $(p, j) \in \text{inv}(\pi^*)$ implies $\pi(i) < \pi(p)$, giving the same contradiction as before.

Suppose $p < i$. Then $(p, i) \notin \text{inv}(\pi)$ implies $\pi(p) < \pi(i)$ and $(p, i) \in \text{inv}(\pi^*)$ implies $\pi(j) < \pi(p)$. This means that $p < i \leq \pi(j) < \pi(p) < \pi(i) \leq j$, and so $p \leq \pi(j) < \pi(p) \leq j$. However, this is impossible since

$$p < i = \min\{t \mid 1 \leq t \leq \pi(j) < \pi(t) \leq j\}.$$

Suppose $j < p$. Then $(j, p) \notin \text{inv}(\pi)$ implies $\pi(j) < \pi(p)$ and $(j, p) \in \text{inv}(\pi^*)$ implies $\pi(i) > \pi(p)$. Furthermore, $\pi(p) \geq p$ since $p > j = \max L(\pi)$. This means that $p \leq \pi(p) < \pi(i) \leq j$ which contradicts our initial assumption that $j < p$.

Hence, there are no elements in $\text{inv}(\pi^*)$ that are not in $\text{inv}(\pi)$, and so $\text{inv}(\pi^*)$ is a subset of $\text{inv}(\pi)$. It is a strict subset since (i, j) is removed. Hence, $\ell(\pi^*) < \ell(\pi)$. $\square$

Now we are ready to prove the main result 3.1.

Proof. We will induct on $\ell(\pi)$. The base case $\ell(\pi) = 1$ is trivial (π is a transposition). Let $\ell \geq 2$ and let π be an n -permutation with $\ell(\pi) = \ell$.

By Lemma 3.3, there are i, j such that $i \leq \pi(j) < \pi(i) \leq j$ and $\ell(\pi^*) < \ell(\pi)$ for $\pi^* = \pi \circ \tau_{ij}$.

By the inductive assumption, there is a good collection $C^* = \{\tau_i\}_{i=1}^k$ that can be ordered so that $\pi^* = \tau_1 \circ \dots \circ \tau_k$ and the sequence of lengths

$$\ell(\tau_1) < \ell(\tau_1 \circ \tau_2) < \dots < \ell(\tau_1 \circ \dots \circ \tau_k)$$

is strictly increasing, and is such that $S_{C^*} = S(\pi^*)$ and $L_{C^*} = L(\pi^*)$. Note that $i < \pi(i)$ and $i \leq \pi^*(i) = \pi(j)$, and so $i \in S(\pi)$ and $i \notin L(\pi^*)$. Similarly, we have $j > \pi(j)$ and $j \geq \pi^*(j) = \pi(i)$, and so $j \in L(\pi)$ and $j \notin S(\pi^*)$. This means that $S(\pi) = S(\pi^*) \cup \{i\}$ and $L(\pi) = L(\pi^*) \cup \{j\}$. Hence, the collection $C = C^* \cup \{\tau_{ij}\}$

with $\tau_{k+1} = \tau_{ij}$ is such that $\pi = \tau_1 \circ \dots \circ \tau_{k+1}$, the sequence of lengths

$$\ell(\tau_1) < \ell(\tau_1 \circ \tau_2) < \dots < \ell(\tau_1 \circ \dots \circ \tau_{k+1})$$

is strictly increasing, and is such that $S_C = S(\pi)$ and $L_C = L(\pi)$. $\square$